MILITARY SPECIFICATION

FIXTURE, CABLE MAINTENANCE

MX-9516/ASQ-81(V)

This specification has been approved
by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - The MX-9516/ASQ-81(V) Cable Maintenance Fixture, herein after referred to as the equipment, covered by this specification shall be utilized for organizational" level use and shall facilitate testing and maintenance of the RL-305A/ASQ-81(V) Reeling Machine Magnetic Detector Launching.

1.2 Classification - The equipment covered by this specification shall consist of the following item:

Type Designation	Applicable Paragraph
Fixture, Cable Maintenance MX-9516/ASQ-81 (V)	3.5.1
Case, Cable Maintenance Fixture	3.5.2

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in 6.9.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the data of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-U-508B	Wiring; Aircraft, Installation of
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

MI L-M-7793	Meter, Time Totalizing
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-T-21200	Test Equipment for Use with Electronic and Fire Control Systems, General Specification for
MI L-STD-461	Electronic Interference Characteristics, Requirements for Equipment
MIL-STD-701	Preferred and Guidance Lists of Semiconductor Devices
MS17322	Meter, Time Totalizing l15 Volt,400-Cycle, Rev. B
MIL-STD-756	Reliability Predication

PUBLICATIONS

Naval Air Systems Command

AR- 5	Microelectronic Devices Used in Avionic Equipment, Procedures for Selection and Approval of
MIL-HDBK-217	Reliability Stress and Failure Rate Data for Electronic Equipment

2.1.1 Availability of Documents - When requesting specifications, standards, drawings and publi cations refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, Code 105, 5801 Tabor Avenue, Philadelphia, Pennsylvania.

3. REQUIREMENTS

3.1 Preproduction - This specification makes provision for preproduction testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

3.2 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials requirements shall conform to MIL-T-21200.

(2) Nonrepairable subassemblies, as outlined in MIL-T-21200, Shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonrepairable subassemblies must be reliable (See 6.4).

(3) When previously produced models of the equipment did not use nonrepairable subassemblies, the design shall not be changed to employ nonrepairable assemblies without the approval of the procuring activity.

3.2.1 Nonstandard Parts and Materials Approval - Approval for the use of nonstandard parts and materials other than microelectronic devices shall be obtained as outlined in MIL-T-21200.

3.2.2 Electronic Devices - Electron tubes and semiconductor devices shall be in accordance with MIL-T-21200 or MIL-STD-454, Req. 30.

3.2.3 Modules, Maintenance - The electronic portions of the equipment shall be divided into maintenance modules as specified In AR-10 (See MIL-S-23603) . Modules shall normally be considered repairable.

3.3 Design and Construction - The equipment shall conform with all the applicable requirements of MIL-T-21200 for design, construction and workmanship except as otherwise specified herein,

3.3.1 Total Weight - The total weight of the equipment shall be a minimum consistent wit, good design and shall not exceed 90 pounds including case.

3.3.2 Reliability

3.3.2.? Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 1,000 hours without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.3.2.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts.

3.3.2.3 Reliability in Mean Time Between Failure (MTBF) - The equipment shall have a minimum of 5,000 hours mean (operating) time between failures, This requirement shall be verified by mathematical analysis in accordance with the provisions of MIL-STD-756 and MIL-HDBK-217.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

3.3.3 Time Totalizing Meter - The equipment shall have a time totalizing meter in accordance with MIL-STD-454, Req. 57.

3.3.4 Cables and Connectors - The equipment shall provide for the use of cables and connectors In accordance with MIL-T-21200.

3.3.6 Wiring (Internal) - Internal wiring shall be in accordance with the requirements o

3.3.6 Wiring (External) - External wiring shall be in accordance with the requirements of MIL-W-5088.

3.3.7 lnterchangeability - The equipment shall meet the inter-changeability requirements as defined in MIL-T-21200.

3.3.8 Interference Control - The generation of radio interference by the equipment shall be controlled within the limits of MIL-STD-461 to the extent described herein.

3.3.8.? Tests required shall be as defined for Class 11 B equipment by Table II of MIL-STD-461A except that electromagnetic and E-field emissions shall not exceed the limits established by MIL-STD-461A for Class II B equip-ment by more than 30 dB.

3.3.9 Maintenance Provisions and Field Testing - Provisions for maintenance snail be as specified in MIL-T-21200. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.3.10 Nomenclature and Nameplates - Nomenclature assignment and namplate approval for equipment identification shall be in accordance with MIL-N-18307.

3.3.11 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests:

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient
Input power voltage	115 ± 1.0 VAC (line to neutral), 3 Phase, 400 ± 5 Hz

3.3.12 Service Conditions - The equipment shall operate satisfactorily under any of the environment service conditions or reasonable combination of these conditions as specified in MIL-T-21200 for Class 2 equipment, except as modified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

3.3.13 Warm-Up Time - The time required for the equipment to warm up prior to operations shall be kept to a minimum and shall not exceed 5 minutes under standard conditions and 10 minutes at extreme service conditions.

3.3.14 Primary Input Power Requirements - The equipment shall be capable of satisfactory operation when supplied input power having characteristics as defined in MIL-T-2l20O. The power required shall be of the following type and shall not exceed the specified amounts.

AC Power, 3 Phase, 115 V, 380-420 Hz, 525 VA

3.4 Performance - The performance of the equipment shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions. When reduced performance requirements for service conditions are acceptable, such variation' will be specified.

3.5 Detail Requirements

3.5.1 Fixture, Cable Maintenance MX-9516/ASQ-81(V) - The equipment shall meet thu following requirements:

3.5.1.1 Function - The function of the equipment is to aid removal or installation and inspection of the tow cable on the RL-305A/ASQ-81(V) Reeling Machine without degrading the cable.

3.5.1.2 Maximum Form Factor - 13.5 inches wide by 10.5 inches high by 31.5 Inches long, not Including case.

3.5.1.3 Weight - The weight of the test set shall not exceed 45 pounds, not including case.

3.5.1,4 Finish - Hard Anodize per MIL-A-8625.

3.5.1.: Contents - The equipment shall contain the following functional circuits or assemb ies:

(1) TorqueMotor- The equipment shall contain a D.C. torque motor and gear train which drives the cable drum with sufficient torque to overcome the cable undertension switch of the Reeling Machine RL-305A/ASQ-81(V) while reeling cable out/in at the slow speed reeling rate and at stall.

(2) Displays - The equipment shall contain a POWER ON indicator light located adjacent to the POWER ON/OFF switch and a time totalizing meter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

(3) Motor Control Circuit - The equipment shall contain a power supply, current regulator and control circuits to convert 3 phase, 400 Hz, 115/200 VAC to a D.C. voltage for operating the cable maintenance fixture.

(4) Levell Wind Mechanism- The equipment shall include a level wind mechanism to provide even wrapping of the cable on the drum.

(5) Cable Drum- The equipment shall include an easily removable drum for cable wrapping. The drum shall nave grooves to aid even cable wrapping and provide more even support of the cable.

3.5.1.6 Controls - The equipment shall contain a POWER ON/OFF switch as its single control.

3.5.1.7 Electrical Connections - Connections to external circuits shall be provided as follows:

Reference Designation	Receptacle Type	Function
W1P;	MS3126F14-12P	Input Power and Control

3.5.2 Case, Cable Maintenance Fixture - The equipment shall be provided with a combination case which shall conform to MIL-T-21200.

3.5.2.: Maximum Size . 17.5 inches wide by 11.5 inches high by 36.5 inches long including hardware and protective devices.

3.5.2.: Material - Aluminum

3.5.2.; Finish - Painted yellow, color 13538 per MIL-STD-595 and finish per MIL-T-21200.

3.5.2.4 Weight - The weight of the case shall not exceed 45 pounds.

3.6 Design Date - to data is required by this specification (other than ports accompanying samples submitted for testing) or applicable documents, unless specified in the contract or order (see paragraph 6.7).

3.7 Workmanship - Workmanship shall be in accordance with MIL-STD-454, Req. 9.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order. the supplier is responsible for the performance of a 1 inspection requirements as-specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the Specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

(1) Preproduction Tests
(2) Acceptance Tests

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.4. The government inspector and the procuring activity shall be advised when tests are . be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services or a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary y t e procuring activity to determine that the equipment meets all the requirements of this specification and the contract. Preproduction test shall include environmental and interference tests in accordance with the requirements of MIL-T-5422 and MIL-STD-461 to the limits specified in MIL-T-21200,

4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prioer to the approval of the preproduction sample is at the contractor’s own risk, The approved preproduction sample shall be retained by THE contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment sahll be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of non-compliance with the above shall *constitute* cause for rejection and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing snail be under the supervision of the government Inspector. Contractors not having testing facilities satisfactory to the procuring activity snail engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests snail consist of the following:

(1) Individual Tests
(2) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product
(2) Operational Test
(3) Manufacturing Run in Test

4.3.1.′ Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

4.3.1.- Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

4.3.1.: Manufacturing Run In Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. The equipment shall be attached to the RL-3G5A/ASQ-81(V) Reeling Machine and power applied to the cable maintenance fixture for a period of 6 hours. The reeling machine shall be cycled (OUT to cable limit and IN to speed-change point) a minimum of 9 times with a minimum of 15 minutes OFF between each half cycle. . failure shall be defined as anything which causes malfunctioning of the cable maintenance fixture. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 2 g's

The equipment shall be vibrated for a period of 10 minutes prior to the beginning of the 6-hour period of operation. The direction of vibration should be vertical to the normal mounting plane. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitio" failures occur, corrective action shall be taken to eliminate this defect f-em future equipment. A record shall be kept of all failures. The 6-hours period specified above may be composed of two 3-hour periods to conform with standar working hours.

4.3.2 Special Tests - Special tests shall be conducted on a quantity of equipments or the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control Procuring activity shall make determination of need for conducting special tests,

4.3.2. Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) Or an early equipment after engineering or material change.

(2) Whenever failure reports or ether information indicate additional tests are required. (This will be determined by the procuring activity).

4.3.2.- Scope of Tests - Special tests shall consist of such tests as approved t. the procuring activity Test procedures previously approved for the preproduction tess shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

4.3.3 Equipment Failures - Should a failure occur during either the individual tests or special tests, the following action shall be taken:

(1) Determine the cause of The failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Determine appropriate corrective action necessary to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonsable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Test Procedures - The procedures used for conducting pre-production tests and acceptance tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

4.5 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shell resubmit the equipment for acceptance.

4.6 Presubmission Testing - No item, part of complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

5. PREPARATION FOR DELIVERY

5.1 Genera? - All major units and part of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-E-17555.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended Use - AN/ASQ-81(V)-2 Magnetic Detecting Set Maintenance.

6.2 Performance Characteristics - Nominal performance characteristics shall be determined at input voltage of 27 + 0.5 VDC and 115 + 1.0 VAC as applicable. (See 3,3.11). Limiting performance characteristics shall be determined at the highest and lowest values of specified input voltages. (See 4.3.2.1). These characteristics shall be used when testing the equipment for compliance with the minimum acceptance standard of performance.

6.3 Performance Objectives - Minimum size and weight. simplicity of operation, ease of mai ntenance and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size, weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-T-21200, it is desired that their use be investiagated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accompanied by complete supporting information.

6.4 AS a general rule, nonrepairable subassemblies should be encapsulated or hermetically-sealed. The number of connections interanl to the subassembly should be held to a minimum. Detail parts tolerances and ratings should be so selected that the life of the subassembly is greater than that of a simalar repairable one. With few exceptions (such as high voltage power supplies), the non-repairable subassembly should evidence a mean-time-to-failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.5 Precedence of Documents - When the requirements of the contract. this speificatlon or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(i) Contract - The contract shall have precedence over any specifications.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from supsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

6.6 The parentheses (*), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature ‘n accordance with 3.3.10. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

6.7 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and date of this specification.

(2) Selection of applicable levels of packing and packaging (see 5.1).

(3) Data requirements (see 3.6).

6.8 Asterisk - In specification revisions and superseding amendments an asterisk preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the asterisk notations and relationship to the last Previous issue.

6.9 Associated Equipment - The equipment shall operate with the followingassociated equipment

AN/ASQ-81A(V)-2 Magnetic Detecting Set

Project No. 4931-N134

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

Naval Air Systems Command
Washington, D.C. 20361

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE NAVY



NAVAL AIR SYSTEMS COMMAND(AIR-52021)
DEPARTMENT OF THE NAVY
WASHINGTON, D.C. 20361

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

SPECIFICATION DIGEST

Form Approved
Budget Bureau No. 22-R355

MIL-F-81595(AS) FIXTURE, [illegible] MX-9516/ASQ-81(V)

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

DIRECT GOVERNMENT CONTRACT SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T06:29Z
Check the source to verify that this is the current version before use.